Technical requirements — AwkBeats Prep Station

Document control

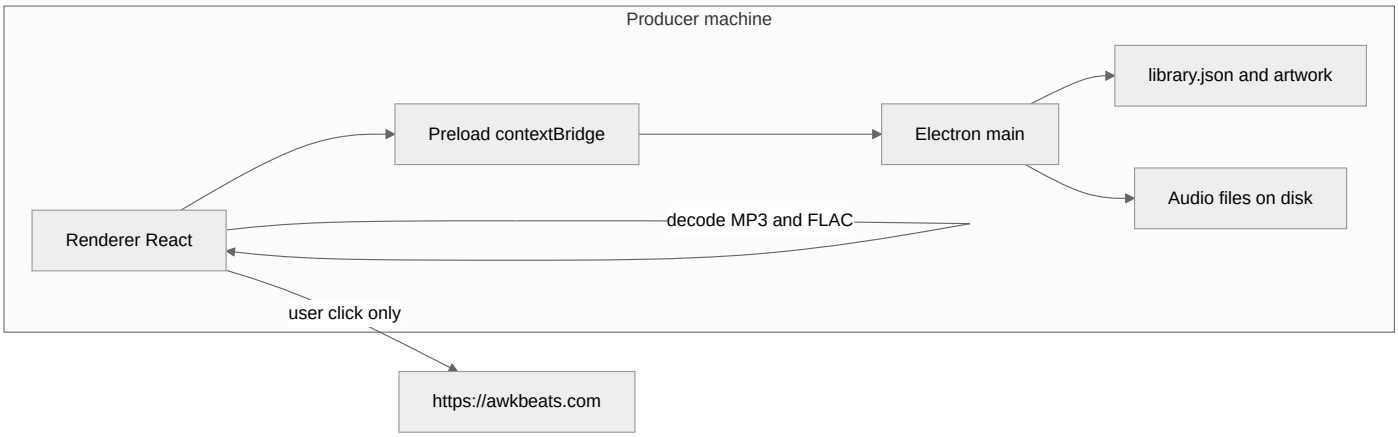
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Operator	Awkward J (legal entity UNCONFIRMED)

Table of contents

1. Architecture
2. Tech stack
3. Repository structure
4. Environments
5. Configuration and secrets
6. Third-party integrations
7. Security
8. Privacy
9. Performance
10. Accessibility
11. SEO
12. Build and deploy
13. Testing
14. Monitoring
15. Browser and device support
16. Technical debt
17. Glossary
18. Open questions
19. Changelog
20. Sources

1. Architecture

The app is a single Electron process plus one sandboxed renderer. There is no application server.



Process split. The renderer never reads the filesystem directly. The main process owns the library and the tag writer.

A request (an IPC call) flows like this. The renderer calls `window.prep`, which is the object exposed by `electron/preload.ts`. The preload uses `ipcRenderer.invoke` with a fixed channel name. `electron/main.ts` handles that channel, calls a class in `src/core`, and returns JSON-serializable data. Audio bytes for MP3 and FLAC analysis travel back to the renderer because Chromium already contains those decoders. WAV and AIFF are decoded in the main process with `PcmDecoder`.

The `beat://` scheme is registered as privileged before `app.ready`. `beat://audio/<id>` and `beat://cover/<id>` are handled with `net.fetch(pathToFileURL(...))` so the renderer can play audio and show art without a raw `file://` path in the page.

2. Tech stack

Versions are the ones installed for the 1.0.0 lockfile.

Layer	Technology	Version	Why	Alternatives rejected
Language	TypeScript	5.9.3	Shared types between main and renderer	Plain JS
Desktop shell	Electron	33.4.11	Mature installers, Chromium decoders for MP3/FLAC	Tauri (no Chromium decoder; would need another permissive decoder)
UI	React	18.3.1	Component model for the three-pane window	No router; one window
Styling	CSS file	`src/renderer/styles.css`	Small surface, no utility framework	Tailwind not needed
Fonts	IBM Plex Sans and Mono via fontsource	5.2.x	OFL, self-hosted	System fonts only would drift between macOS and Windows
Renderer build	Vite	6.4.5	Fast dev server	webpack
Main and preload build	esbuild	0.24.2	Bundle CJS, externalize electron and music-metadata	tsc emit
Test	Vitest	2.1.9	Same TS path aliases	Jest
Package	electron-builder	25.1.8	dmg and NSIS	Manual zip
Tags write	node-id3	0.2.9	MIT ID3v2 writer	Any GPL tagger
Tags read	music-metadata	7.14.0	MIT, CJS	—
Key and BPM	Original TypeScript	—	Permissive, testable	essentia.js (AGPL), aubio (GPL)

Layer	Technology	Version	Why	Alternatives rejected
Catalog	JSON file	schema version 1	No native module	SQLite / sql.js (wasm packaging deferred)
CI	GitHub Actions	checkout/setup-node/upload-artifact v4	macOS and Windows runners	Local cross-compile

Node in CI is 22. The development machine that produced this tree used Node 22.23.3 and npm 10.9.9.

State management is React `useState` in `src/renderer/App.tsx`. There is no Redux or router.

Linting: `tsc --noEmit` is the type gate. There is no ESLint config in v1.

3. Repository structure

```
electron/main.ts          window, menu, IPC, beat:// protocol
electron/preload.ts       PrepApi bridge
src/core/analysis/        Fft, BpmDetector, KeyDetector, Camelot, BpmMath, TrackAnalyzer
src/core/audio/           PcmDecoder, ReferenceAudio (tests and generators)
src/core/library/         LibraryStore, Importer, BeatRules, CsvExport, SquareCrop, types
src/core/tags/            TagReader, TagWriter, TagCommit, TagTypes
src/renderer/             React UI, styles, index.html CSP
tests/                    bpm-key and library-tags
tests/fixtures/silence.flac
scripts/dev.mjs           Vite + esbuild watch + Electron
scripts/check-licenses.mjs
scripts/seed-demo.ts      local demo library, not shipped in the installer
build/icon.png, logo.png, icon.ico, icon.icns, icons/
electron-builder.yml
.github/workflows/installers.yml
docs/                     this documentation set
```

`dist/` and `release/` are build outputs and are gitignored.

4. Environments

Environment	URL	How it runs	Difference
Local dev	`http://localhost:5173` inside Electron, set as `VITE_DEV_SERVER_URL`	`npm run dev`	Vite HMR. CSP allows that origin.
Local production build	`file://` to `dist/renderer/index.html`	`npm run build && npm start`	No dev server.
Installer	none until the owner distributes the artifact	`npm run dist:mac` or `dist:win`, or CI	asar archive. Unsigned.
Staging	Not applicable	No staging host	—
Hosted production	Not applicable	The app does not deploy to a server	—

`AWKBEATS_USER_DATA` points the library at a chosen folder. Unset, Electron uses its default userData path for app id `com.awkbeats.prepstation`.

5. Configuration and secrets

No secret is required to build or run the unsigned app. Signing secrets are owner-supplied later and must never be committed.

Name	Purpose	Required	Default	Where set	Where read	Rotation
`AWKBEATS_USER_DATA`	Override library directory	Optional	Electron userData	Process environment	`electron/main.ts` before ready	Change the path; move `library.json` and `artwork/` together
`VITE_DEV_SERVER_URL`	Dev renderer origin	Dev only	Unset in production	`scripts/dev.mjs`	`electron/main.ts` `createWindow`	Restart dev
`CSC_LINK`	Base64 Apple or Windows signing cert	Optional, signing only	Unset	GitHub Actions secret	electron-builder, not application code	Revoke the cert with the vendor, replace the secret
`CSC_KEY_PASSWORD`	Password for `CSC_LINK`	With `CSC_LINK`	Unset	GitHub Actions secret	electron-builder	Same as the cert
`WIN_CSC_LINK` / `WIN_CSC_KEY_PASSWORD`	Windows-specific override	Optional	Unset	GitHub Actions secret	electron-builder	Same
`APPLE_API_KEY`, `APPLE_API_KEY_ID`, `APPLE_API_ISSUER`, `APPLE_TEAM_ID`	Notarization	Optional	Unset	GitHub Actions secret	electron-builder notarize	Rotate the API key in App Store Connect
`CSC_IDENTITY_AUTO_DISCOVERY`	Stop the builder from grabbing a random local cert	Set to `false` in CI	false in the workflow	`.github/workflows/installers.yml`	electron-builder	Leave false until a signing job exists
`GITHUB_TOKEN`	Default Actions token	Present on Actions	—	GitHub	Not read by app code. `publish: null` and `--publish never` stop electron-builder from creating a Release	GitHub rotates it

There is no `.env` file. Do not add signing material to one.

6. Third-party integrations

No payment, email, auth, analytics, storage, CDN, or AI API is called.

Integration	Account	Endpoints	Auth	Failure
awkbeats.com	The owner's site. Account ID UNCONFIRMED.	`https://awkbeats.com` via `shell.openExternal` when the brand control is activated	None	OS browser handles failure. The library still works.
GitHub Actions	Repo Awkward8787/awkbeats-prep-station	Workflow artifacts only	GitHub permissions: `contents: read`	Installer job fails if `release/*.dmg` or `release/*.exe` is missing
music-metadata, node-id3	npm packages, not accounts	Local function calls	None	Tag read/write throws; UI shows the tag-failed message or the import failure count

Sandbox versus production: not applicable. There is one build.

7. Security

Authentication: none. The person at the keyboard is the user. The library is as private as the OS user account.

Authorization: IPC channels are a fixed list in `electron/preload.ts`. The renderer cannot pass a channel name of its own choosing through the bridge. Paths come from the file dialog or from `webUtils.getPathForFile` on a dropped file.

Input validation (`BeatValidation`):

- ISRC: empty, or `/^[A-Z]{2}[A-Z0-9]{3}\d{7}$/` after removing hyphens and spaces and uppercasing. Stored canonical, 12 characters.
- BPM override: 40 through 300 inclusive (`BpmMath.inRange`).
- Tags: trim, collapse whitespace, case-insensitive de-dupe, max 24 tags, 32 characters each.
- Status: one of `idea`, `in_progress`, `ready`, `leased`, `sold`.
- Comment max 4000 characters. Copyright max 400.
- Audio import: extension allow-list, size greater than 0 and at most 2 GiB, skip dotfiles.
- Cover export size: 1500 or 3000 only.

CSRF and CORS: not applicable to a local window. The `beat` scheme sets `corsEnabled` so the renderer can fetch it. Navigation: `setWindowOpenHandler` returns deny. `will-navigate` allows only the dev server origin and `file:`.

Rate limiting: not applicable. There is no public endpoint.

CSP in `src/renderer/index.html`:

```
default-src 'self' beat:;
script-src 'self';
style-src 'self' 'unsafe-inline';
img-src 'self' data: blob: beat:;
media-src 'self' blob: beat:;
connect-src 'self' beat: http://localhost:5173 ws://localhost:5173;
```

`style-src 'unsafe-inline'` is there because the cover well sets a background image in an element style and the crop stage positions an image with inline coordinates. There is no HSTS header because the document is not served over HTTPS. The brand site is opened in the system browser.

Secret storage: none in the app. Dependency audit: `npm run licenses` fails on GPL-family licenses in production packages. That check passed on the 1.0.0 tree. File permissions: the library directory inherits the user's umask. Known accepted risk: an unsigned binary will be flagged by Gatekeeper and SmartScreen. Another accepted risk: `connect-src` includes localhost so the dev build can load Vite. The production window does not start a server, but the CSP text is the same HTML file.

8. Privacy

Data	Where	Retention	Third party
Audio the producer imports	Original path on disk. The app does not copy the audio into userData.	Until the producer deletes the file	None
Catalog	`library.json` in userData	Until the producer removes beats or deletes the folder	None
Cover JPEGs	`userData/artwork/<id>.jpg`	Deleted when the beat is removed from the library	None

Data	Where	Retention	Third party
Cookies, localStorage, IndexedDB	Not used	—	—

`library.json` can contain file paths, which may include the OS username. That file stays on the machine. The app does not ship a logger that uploads.

9. Performance

Production renderer build measured on this tree:

Asset	Size
`dist/renderer/assets/index-*.js`	190.62 kB (Vite, gzip 61.32 kB)
`dist/renderer/assets/index-*.css`	16.59 kB (Vite, gzip 3.64 kB)
IBM Plex woff2 subsets	about 8–16 KB each, weights imported: sans 400/500/600, mono 500
`dist/main/index.js`	668 KB (node-id3 bundled; music-metadata external)
`dist/preload/index.js`	4 KB

Code splitting: one renderer chunk. The window is one screen. Caching headers: not applicable (file load). Images: covers are JPEG written by the crop dialog at quality 0.92, or the original sidecar bytes. Fonts are self-hosted woff2. No CDN.

Analysis reads at most the first 90 seconds. Onset detection uses hop 32. Key detection uses FFT size 16384 and hop 8192. The waveform stored on the beat is 160 peaks, each 0–1 with three decimal places.

10. Accessibility

- The search field has a label ("Search the library") that is visually clipped with the `.sr` class and tied to `library-search`. Other fields have visible labels.
- Focus: `:focus-visible` is a 2px solid `#f3d7a4` ring, 2px offset.
- Keyboard: listed in `ShortcutsDialog` and implemented in `App.tsx`. Status keys 1–5 map onto `BEAT_STATUSES` order. Shortcuts do not fire while an input, textarea, or select is focused, except Escape closes a dialog.
- Dialogs close with Escape, Cancel, or Close. Focus is not trapped in a WAI-ARIA modal loop in v1 (see debt).
- Contrast: see ABP-DESIGN. Body text on `#0c0d10` is 17.1:1. Muted `#938d85` on `#2a2418` is 4.50:1, which meets WCAG AA for normal text.
- Reduced motion: all transitions and animations are disabled.
- The play button is 36×36 CSS pixels. Table rows are 44px tall. Filter rows are about 32px, under a 44px target. Noted as debt.
- Color is not the only status signal. The status word is also printed.
- There is no skip link. The window has no repeated web header to skip.

11. SEO and sharing

Not applicable. The app has no public routes, sitemap, robots.txt, canonical URL, or Open Graph tags. The HTML title is "AwkBeats Prep Station". The installer icons are `build/icon.png` (1024) and `build/icon.ico`.

12. Build and deploy

Commands:

```
npm ci
npm test
npm run typecheck
npm run licenses
npm run build
```

`npm run build` runs esbuild for `electron/main.ts` and `electron/preload.ts` (CJS, electron external, music-metadata external on main) and `vite build` with root `src/renderer`, `base: './'`, outDir `dist/renderer`.

Installers:

```
npm run dist:mac
npm run dist:win
```

Both pass `--publish never`. CI (`.github/workflows/installers.yml`) runs test, typecheck, and licenses on `ubuntu-latest`, then `dist:mac` on `macos-latest` and `dist:win` on `windows-latest`. Artifacts:

- `AwkBeats-Prep-Station-macOS` from `release/*.dmg`
- `AwkBeats-Prep-Station-Windows` from `release/*.exe`

`if-no-files-found: error`. Permissions are `contents: read`. `publish: null` in `electron-builder.yml`.

There is no cache to purge and no server to roll back. Rollback of an installed copy is the OS uninstall plus installing the previous artifact. Rollback of the repo is `git revert`.

Post-build check: `release/` contains the dmg or exe; the app launches; the unit tests still match known tempos.

13. Testing

Framework: Vitest 2.1.9, `vitest.config.ts`, timeout 30 seconds, alias `@core` to `src/core`.

`tests/bpm-key.test.ts` asserts:

- FFT magnitude peak location.
- Camelot parse and format, including F# minor = 11A and flats mapping onto sharps.
- `BpmMath` half, double, range, and working tempo.
- Click tracks at 72.5, 90, 96, 120, 128.05, 140.25, 150, and 174 BPM, each within 0.01.
- Noise bursts at 100 BPM within 0.01.
- Soft pulses at 110 BPM within 0.01 and confidence greater than 0.8.
- An 80 BPM click is not reported as 160.
- All 12 major and 12 minor keys on `ReferenceAudio.keyedTone`.
- A drum loop at 96 BPM in F# minor returns 96.00 and `11A`.

`tests/library-tags.test.ts` asserts PCM round-trip error under 1/30000; MP3, WAV, AIFF, and FLAC tag round-trip including decimal BPM, Camelot, tags, ISRC, and artwork; WAV audio samples unchanged after a tag rewrite; import, duplicate skip, lock surviving re-analysis, filters, CSV, ISRC validation, crop math, and batch status.

Run: `npm test`. CI status: the workflow file is in the repo. The run URL is the Actions run for the shipping commit (see ABP-PLAN).

Manual QA: launch, import, read key and BPM, edit metadata, open crop at 3000, write tags, export CSV. Screenshots of the library, the analysis card, the metadata editor, and the crop dialog were taken on a Linux Electron window during development. macOS and Windows GUI launch is not part of the workflow. The runners prove the installer files exist.

Known gaps: no automated UI test. No test that Chromium's MP3 decoder matches a fixture BPM. The detector itself is tested on PCM. No installer GUI test on this Linux tree.

14. Monitoring and logging

Electron and Chromium write their default stderr. The app does not log file contents, ISRC lists, or keystrokes to a server. A packaged app's logs stay in the OS console. Alerts and uptime checks are not applicable. There is no service to ping.

15. Browser and device support

Target	Support
macOS arm64 and x64	dmg via electron-builder. Exact minimum macOS is whatever Electron 33.4.11 declares (UNCONFIRMED as a marketing string).
Windows x64	NSIS exe. Per-user install, directory changeable, not one-click.
Windows arm64, Linux packages	Not built
Web browsers	Not a target. The renderer is Chromium inside Electron 33.
Phone	Not a target. Below 980px CSS pixels the layout collapses to two columns. That is for a small desktop window.

16. Technical debt and known issues

Item	Severity	Proposed fix
Unsigned installers	High for distribution	Owner certificates, then a signing job.
Dialogs are not focus-trapped	Low	Add a focus trap when a modal opens.
Some hit targets under 44px	Low	Increase `.status-item` padding if an audit requires a 44px target.
`style-src 'unsafe-inline'`	Low	Move cover and crop positioning to CSS variables set from classes.
Catalog is one JSON rewrite per edit	Low until thousands of beats	Migrate to SQLite with the same `Beat` fields.
music-metadata parses BPM as an integer	Low, already handled	`TagReader.readNative` keeps the decimal. Do not remove that path.
Half-time of a mix without a clear kick can still be the tempo the producer wants	Medium, product	The UI offers half, double, and a typed BPM.
`concurrently` and `wait-on` are unused devDependencies	Low	Remove on a later cleanup.
React and fontsource are production dependencies even though Vite bundles them	Low	Move to devDependencies to slim the asar. music-metadata must stay a production dependency. node-id3 is bundled into the main file.
Search label is screen-reader only	Low	The field is the only text input in the top bar. Add a visible label if a later audit wants it.

17. Glossary

Term	Meaning
IPC	Electron `ipcMain.handle` / `ipcRenderer.invoke` channels listed in ABP-DATA.
asar	Electron archive of `dist/**` and `package.json`.
NSIS	The Windows installer technology electron-builder uses.
PrepApi	The TypeScript type of `window.prep` in `electron/preload.ts`.

18. Open questions

ID	Question	Owner	Blocked
Q-001	Legal entity for the Developer ID certificate	Awkward J	Signed macOS build
Q-007	Which Windows certificate vendor, OV or EV?	Awkward J	SmartScreen reputation. EV is the one that trusts immediately.
Q-008	Confirm Electron's declared minimum macOS for the 33.4.11 binary before publishing a support line	Awkward J	Support page, which does not exist yet

19. Changelog

Date	Version	Change	Why
2026-10-21	1.0	Initial TRD for the 1.0.0 Electron tree.	Owner build brief. Muted text color raised to `#938d85` so small labels meet WCAG AA.
2026-10-21	1.1	Installer icons are `build/icon.icns`, `build/icon.ico`, and `build/icons/`.	Owner supplied the iOS icon.

20. Sources

- `package.json`, installed versions listed in section 2.
- `electron/main.ts`, `electron/preload.ts`, `electron-builder.yml`.
- `src/renderer/index.html`, `vite.config.ts`, `vitest.config.ts`, `tsconfig.json`.
- `.github/workflows/installers.yml`, `scripts/check-licenses.mjs`, `scripts/dev.mjs`.
- `tests/bpm-key.test.ts`, `tests/library-tags.test.ts`.
- Bundle sizes from `du` on `dist/` after `npm run build` on 2026-10-21.